Final Thesis Project Code

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R Markdown

```
knitr::opts_chunk$set(
  fig.width = 24,      # Figure width
  fig.height = 16,     # Figure height
  echo = TRUE,
  warning = FALSE,     # Suppress warnings in output
  message = FALSE      # Suppress messages in output
)

# Set default CRAN mirror (fix contrib.url error)
options(repos = c(CRAN = "https://cran.rstudio.com"))

install.packages("readxl", dependencies = TRUE)

## package 'readxl' successfully unpacked and MD5 sums checked

##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages

install.packages("dplyr", dependencies = TRUE)

##
##   There are binary versions available but the source versions are later:
##       binary source needs_compilation
## rlang  1.1.5  1.1.7              TRUE
## vctrs  0.6.5  0.7.2              TRUE
## dplyr  1.1.4  1.2.0              TRUE

install.packages("Benchmarking", dependencies = TRUE)

## package 'Benchmarking' successfully unpacked and MD5 sums checked
```

```
##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages
```

```r
install.packages("truncreg",dependencies = TRUE )
```

```
## package 'truncreg' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages
```

```r
install.packages("cellranger", dependencies = TRUE)
```

```
## package 'cellranger' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages
```

```r
install.packages("ggplot2", dependencies = TRUE)
```

```
##
##   There is a binary version available but the source version is later:
##         binary source needs_compilation
## ggplot2  3.5.1  4.0.2             FALSE
```

```r
install.packages("tidyr", dependencies = TRUE)
```

```
##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## tidyr  1.3.1  1.3.2              TRUE
```

```r
install.packages("AER", dependencies = TRUE)
```

```
##
##   There is a binary version available but the source version is later:
##     binary source needs_compilation
## AER 1.2-14 1.2-16             FALSE
```

```r
install.packages("stargazer", dependencies = TRUE)
```

```
## package 'stargazer' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages
```

```r
install.packages("psych", dependencies = TRUE)
```

```
##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## psych  2.5.3  2.6.3             FALSE
```

```
install.packages("knitr", dependencies = TRUE)

##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## knitr   1.50   1.51             FALSE

install.packages("gtable", dependencies = TRUE)

## package 'gtable' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\xuanq\AppData\Local\Temp\RtmpiA0K2Q\downloaded_packages

 install.packages("knitr", dependencies = TRUE)

##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## knitr   1.50   1.51             FALSE

install.packages("psych", dependencies = TRUE)

##
##   There is a binary version available but the source version is later:
##       binary source needs_compilation
## psych  2.5.3  2.6.3             FALSE

library(readxl)
library(Benchmarking)
library(dplyr)
library(cellranger)
library(ggplot2)
library(tidyr)
library(psych)
library(knitr)
library(AER)
library(stargazer)
library(truncreg)
library(gridExtra)
library(texreg)
library(xfun)
```

Phase 1: Data Envelopment Analysis (DEA)

This phase calculates Overall Technical Efficiency (OTE) using the CRS model, Pure Technical Efficiency (PTE) using the VRS model, and Scale Efficiency (SE).

#1.1 Data Import and Initial Cleaning for DEA

```r
dea_raw <- read_excel("D:/dea/deaxl/DEA.xlsx", sheet = "Tổng hợp", range =
cell_limits(c(1, 1), c(211, 11)))

cat("--- Raw DEA Data after Import (dea_raw) ---\n")
```

```
## --- Raw DEA Data after Import (dea_raw) ---
```

```r
head(dea_raw, n = 10)
```

```
## # A tibble: 10 × 11
##      No. Firm  `Life insurance?`  Year `DEA Score` `Financial Investment
Income`
##    <dbl> <chr> <chr>             <dbl> <lgl>
<dbl>
##  1     1 AIA   Yes                2016 NA
696829019210
##  2     2 AIA   <NA>               2017 NA
1050484101385
##  3     3 AIA   <NA>               2018 NA
1053897057514
##  4     4 AIA   <NA>               2019 NA
1297311077184
##  5     5 AIA   <NA>               2020 NA
1892257469684
##  6     6 AIA   <NA>               2021 NA
2009330763558
##  7     7 AIA   <NA>               2022 NA
1947466657244
##  8     8 ABIC  No                 2016 NA
63024097982
##  9     9 ABIC  <NA>               2017 NA
80501566490
## 10    10 ABIC  <NA>               2018 NA
96485515420
## # ℹ 5 more variables: `Benefits/ losses incurred` <dbl>, `Total Equity`
<dbl>,
## #   `Total Debt` <dbl>, `Total Operating Exp` <dbl>,
## #   `Dummy (Life_Insurance=1)` <dbl>
```

```r
cat("--- Raw DEA Data after Import (dea_raw) ---\n")
```

```
## --- Raw DEA Data after Import (dea_raw) ---
```

#1.2 Check Dimensions and Column Names of Raw Data

```r
print(dim(dea_raw))
```

```
## [1] 210  11
```

```r
head(dea_raw, n = 10)
```

```
## # A tibble: 10 × 11
##      No. Firm  `Life insurance?`  Year `DEA Score` `Financial Investment
Income`
##    <dbl> <chr> <chr>             <dbl> <lgl>
<dbl>
##  1     1 AIA   Yes                2016 NA
696829019210
##  2     2 AIA   <NA>               2017 NA
1050484101385
##  3     3 AIA   <NA>               2018 NA
1053897057514
##  4     4 AIA   <NA>               2019 NA
1297311077184
##  5     5 AIA   <NA>               2020 NA
1892257469684
##  6     6 AIA   <NA>               2021 NA
2009330763558
##  7     7 AIA   <NA>               2022 NA
1947466657244
##  8     8 ABIC  No                 2016 NA
63024097982
##  9     9 ABIC  <NA>               2017 NA
80501566490
## 10    10 ABIC  <NA>               2018 NA
96485515420
## # ℹ 5 more variables: `Benefits/ losses incurred` <dbl>, `Total Equity`
<dbl>,
## #   `Total Debt` <dbl>, `Total Operating Exp` <dbl>,
## #   `Dummy (Life_Insurance=1)` <dbl>
```

```r
cat("\n--- Column Names in dea_raw ---\n")
```

```
##
## --- Column Names in dea_raw ---
```

```r
print(colnames(dea_raw))
```

```
##  [1] "No."                          "Firm"
##  [3] "Life insurance?"              "Year"
##  [5] "DEA Score"                    "Financial Investment Income"
##  [7] "Benefits/ losses incurred"    "Total Equity"
##  [9] "Total Debt"                   "Total Operating Exp"
## [11] "Dummy (Life_Insurance=1)"
```

#1.3 Clean and Transform DEA Data

```r
# Clean and transform variables for DEA analysis
# Filter out rows with NA values in critical input/output columns and the
firm type column.
# Convert relevant columns to numeric type.
# Create a clear 'Firm_Type' column based on 'Life_Insurance=1'.
```

```r
dea_clean <- dea_raw %>%
 filter(!is.na(`Financial Investment Income`),
        !is.na(`Benefits/ losses incurred`),
        !is.na(`Total Operating Exp`),
        !is.na(`Total Debt`),
        !is.na(`Total Equity`),
        !is.na(`Dummy (Life_Insurance=1)`)) %>%
  mutate(across(c(`Financial Investment Income`, `Benefits/ losses incurred`,
`Total Operating Exp`, `Total Debt`, `Total Equity`, `Dummy
(Life_Insurance=1)`), as.numeric)) %>%
 mutate(Firm_Type = ifelse(`Dummy (Life_Insurance=1)` == 1, "Life", "Non-
Life"))
cat("\n--- Cleaned DEA Data with Firm Type (dea_clean) ---\n")
```

```
##
## --- Cleaned DEA Data with Firm Type (dea_clean) ---
```

```r
head(dea_clean)
```

```
## # A tibble: 6 × 12
##     No. Firm  `Life insurance?`  Year `DEA Score` `Financial Investment
Income`
##   <dbl> <chr> <chr>             <dbl> <lgl>
<dbl>
## 1     1 AIA   Yes                2016 NA
696829019210
## 2     2 AIA   <NA>               2017 NA
1050484101385
## 3     3 AIA   <NA>               2018 NA
1053897057514
## 4     4 AIA   <NA>               2019 NA
1297311077184
## 5     5 AIA   <NA>               2020 NA
1892257469684
## 6     6 AIA   <NA>               2021 NA
2009330763558
## # ℹ 6 more variables: `Benefits/ losses incurred` <dbl>, `Total Equity`
<dbl>,
## #   `Total Debt` <dbl>, `Total Operating Exp` <dbl>,
## #   `Dummy (Life_Insurance=1)` <dbl>, Firm_Type <chr>
```

```r
str(dea_clean)
```

```
## tibble [210 × 12] (S3: tbl_df/tbl/data.frame)
##  $ No.                       : num [1:210] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Firm                      : chr [1:210] "AIA" "AIA" "AIA" "AIA" ...
##  $ Life insurance?           : chr [1:210] "Yes" NA NA NA ...
##  $ Year                      : num [1:210] 2016 2017 2018 2019 2020 ...
##  $ DEA Score                 : logi [1:210] NA NA NA NA NA NA ...
##  $ Financial Investment Income: num [1:210] 6.97e+11 1.05e+12 1.05e+12
1.30e+12 1.89e+12 ...
```

```
##  $ Benefits/ losses incurred  : num [1:210] 2.88e+12 4.45e+12 5.57e+12
6.65e+12 8.47e+12 ...
##  $ Total Equity               : num [1:210] 1.48e+12 3.71e+12 4.17e+12
4.98e+12 6.55e+12 ...
##  $ Total Debt                 : num [1:210] 8.58e+12 1.17e+13 1.46e+13
1.87e+13 2.38e+13 ...
##  $ Total Operating Exp        : num [1:210] 5.14e+12 6.97e+12 8.86e+12
1.11e+13 1.38e+13 ...
##  $ Dummy (Life_Insurance=1)   : num [1:210] 1 1 1 1 1 1 1 0 0 0 ...
##  $ Firm_Type                  : chr [1:210] "Life" "Life" "Life" "Life"
...
```

1.4 Define Inputs and Outputs for DEA

```r
# Split data into Life and Non-Life firms for separate DEA analysis
dea_life_data <- dea_clean %>% filter(Firm_Type == "Life")
dea_nonlife_data <- dea_clean %>% filter(Firm_Type == "Non-Life")

cat("\n--- Dimensions of Life Insurance Data for DEA ---\n")
```

```
##
## --- Dimensions of Life Insurance Data for DEA ---
```

```r
print(dim(dea_life_data))
```

```
## [1] 98 12
```

```r
cat("\n--- Dimensions of Non-Life Insurance Data for DEA ---\n")
```

```
##
## --- Dimensions of Non-Life Insurance Data for DEA ---
```

```r
print(dim(dea_nonlife_data))
```

```
## [1] 112  12
```

```r
# Define inputs and outputs for Life Insurance Firms
inputs_life <- as.matrix(dea_life_data[, c("Total Operating Exp", "Total
Debt", "Total Equity")])
outputs_life <- as.matrix(dea_life_data[, c("Financial Investment Income",
"Benefits/ losses incurred")])

# Define inputs and outputs for Non-Life Insurance Firms
inputs_nonlife <- as.matrix(dea_nonlife_data[, c("Total Operating Exp",
"Total Debt", "Total Equity")])
outputs_nonlife <- as.matrix(dea_nonlife_data[, c("Financial Investment
Income", "Benefits/ losses incurred")])

cat("\n--- Dimensions of Inputs and Outputs Matrices for Life Firms ---\n")
```

```
##
## --- Dimensions of Inputs and Outputs Matrices for Life Firms ---

print(dim(inputs_life))

## [1] 98  3

print(dim(outputs_life))

## [1] 98  2

cat("\n--- Dimensions of Inputs and Outputs Matrices for Non-Life Firms ---
\n")

##
## --- Dimensions of Inputs and Outputs Matrices for Non-Life Firms ---

print(dim(inputs_nonlife))

## [1] 112   3

print(dim(outputs_nonlife))

## [1] 112   2
```

#1.5 Run DEA Models: CRS, VRS, and Calculate Scale Efficiency

```
# --- For Life Insurance Firms ---
life_eff_scores <- data.frame(Firm = dea_life_data$Firm, Year =
dea_life_data$Year, Firm_Type = "Life")
if (nrow(dea_life_data) > 0 && ncol(inputs_life) > 0 && ncol(outputs_life) >
0 && sum(rowSums(inputs_life) > 0) > 0 && sum(rowSums(outputs_life) > 0) > 0)
{
  dea_result_life_crs <- dea(inputs_life, outputs_life, RTS = "crs",
ORIENTATION = "in")
  dea_result_life_vrs <- dea(inputs_life, outputs_life, RTS = "vrs",
ORIENTATION = "in")
  life_eff_scores$OTE <- dea_result_life_crs$eff
  life_eff_scores$PTE <- dea_result_life_vrs$eff
  life_eff_scores$SE <- life_eff_scores$OTE / life_eff_scores$PTE
  cat("\n--- Summary of DEA Results (Life Firms - CRS Model) ---\n")
  summary(dea_result_life_crs)
  cat("\n--- Summary of DEA Results (Life Firms - VRS Model) ---\n")
  summary(dea_result_life_vrs)
} else {
  cat("\nWarning: Insufficient data or invalid inputs/outputs for DEA for
Life Insurance Firms. Efficiency scores will be NA.\n")
  life_eff_scores$OTE <- NA
  life_eff_scores$PTE <- NA
  life_eff_scores$SE <- NA
}
```

```
##
## --- Summary of DEA Results (Life Firms - CRS Model) ---
## Summary of efficiencies
## CRS technology and input orientated efficiency
## Number of firms with efficiency==1 are 16 out of 98
## Mean efficiency: 0.858
## ---
##   Eff range       #    %
##   0.4<= E <0.5    1  1.0
##   0.5<= E <0.6    2  2.0
##   0.6<= E <0.7    9  9.2
##   0.7<= E <0.8   17 17.3
##   0.8<= E <0.9   26 26.5
##   0.9<= E <1     27 27.6
##         E ==1    16 16.3
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.4979  0.7831  0.8796  0.8581  0.9629  1.0000
##
## --- Summary of DEA Results (Life Firms - VRS Model) ---
## Summary of efficiencies
## VRS technology and input orientated efficiency
## Number of firms with efficiency==1 are 26 out of 98
## Mean efficiency: 0.886
## ---
##   Eff range       #    %
##   0.5<= E <0.6    3  3.1
##   0.6<= E <0.7    4  4.1
##   0.7<= E <0.8   16 16.3
##   0.8<= E <0.9   20 20.4
##   0.9<= E <1     29 29.6
##         E ==1    26 26.5
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.5313  0.8050  0.9190  0.8861  1.0000  1.0000
```

```
# --- For Non-Life Insurance Firms ---
nonlife_eff_scores <- data.frame(Firm = dea_nonlife_data$Firm, Year =
dea_nonlife_data$Year, Firm_Type = "Non-Life")
if (nrow(dea_nonlife_data) > 0 && ncol(inputs_nonlife) > 0 &&
ncol(outputs_nonlife) > 0 && sum(rowSums(inputs_nonlife) > 0) > 0 &&
sum(rowSums(outputs_nonlife) > 0) > 0) {
  dea_result_nonlife_crs <- dea(inputs_nonlife, outputs_nonlife, RTS = "crs",
ORIENTATION = "in")
  dea_result_nonlife_vrs <- dea(inputs_nonlife, outputs_nonlife, RTS = "vrs",
ORIENTATION = "in")
  nonlife_eff_scores$OTE <- dea_result_nonlife_crs$eff
  nonlife_eff_scores$PTE <- dea_result_nonlife_vrs$eff
  nonlife_eff_scores$SE <- nonlife_eff_scores$OTE / nonlife_eff_scores$PTE
  cat("\n--- Summary of DEA Results (Non-Life Firms - CRS Model) ---\n")
  summary(dea_result_nonlife_crs)
  cat("\n--- Summary of DEA Results (Non-Life Firms - VRS Model) ---\n")
```

```
  summary(dea_result_nonlife_vrs)
} else {
  cat("\nWarning: Insufficient data or invalid inputs/outputs for DEA for
Non-Life Insurance Firms. Efficiency scores will be NA.\n")
  nonlife_eff_scores$OTE <- NA
  nonlife_eff_scores$PTE <- NA
  nonlife_eff_scores$SE <- NA
}
```

```
##
## --- Summary of DEA Results (Non-Life Firms - CRS Model) ---
## Summary of efficiencies
## CRS technology and input orientated efficiency
## Number of firms with efficiency==1 are 15 out of 112
## Mean efficiency: 0.833
## ---
##   Eff range       #    %
##   0.5<= E <0.6    3  2.7
##   0.6<= E <0.7    6  5.4
##   0.7<= E <0.8   35 31.2
##   0.8<= E <0.9   39 34.8
##   0.9<= E <1     14 12.5
##         E ==1    15 13.4
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.5189  0.7451  0.8369  0.8327  0.9090  1.0000
##
## --- Summary of DEA Results (Non-Life Firms - VRS Model) ---
## Summary of efficiencies
## VRS technology and input orientated efficiency
## Number of firms with efficiency==1 are 29 out of 112
## Mean efficiency: 0.908
## ---
##   Eff range       #    %
##   0.6<= E <0.7    4  3.6
##   0.7<= E <0.8    8  7.1
##   0.8<= E <0.9   38 33.9
##   0.9<= E <1     33 29.5
##         E ==1    29 25.9
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.6663  0.8539  0.9206  0.9077  1.0000  1.0000
```

#1.6 Combine Efficiency Scores (Life + Non-Life)

```
# Combine efficiency scores from both groups
all_eff_scores_detailed <- bind_rows(life_eff_scores, nonlife_eff_scores)

cat("\n--- All Efficiency Scores (Combined, Group-Specific DEA) ---\n")
```

```
##
## --- All Efficiency Scores (Combined, Group-Specific DEA) ---
```

```r
head(all_eff_scores_detailed)
```

```
##   Firm Year Firm_Type       OTE       PTE        SE
## 1  AIA 2016      Life 0.7884149 0.7938695 0.9931290
## 2  AIA 2017      Life 0.8086187 0.8098457 0.9984849
## 3  AIA 2018      Life 0.8102293 0.8115008 0.9984332
## 4  AIA 2019      Life 0.7741838 0.7789558 0.9938739
## 5  AIA 2020      Life 0.7812774 0.7865433 0.9933050
## 6  AIA 2021      Life 0.8904749 0.9085229 0.9801348
```

```r
str(all_eff_scores_detailed)
```

```
## 'data.frame':    210 obs. of  6 variables:
##  $ Firm     : chr  "AIA" "AIA" "AIA" "AIA" ...
##  $ Year     : num  2016 2017 2018 2019 2020 ...
##  $ Firm_Type: chr  "Life" "Life" "Life" "Life" ...
##  $ OTE      : num  0.788 0.809 0.81 0.774 0.781 ...
##  $ PTE      : num  0.794 0.81 0.812 0.779 0.787 ...
##  $ SE       : num  0.993 0.998 0.998 0.994 0.993 ...
```

#1.7 Filter for Valid Efficiency Scores

```r
plot_data_all_eff <- all_eff_scores_detailed %>%
  filter(!is.na(OTE) & OTE >= 0 & OTE <= 1,
         !is.na(PTE) & PTE >= 0 & PTE <= 1,
         !is.na(SE) & SE >= 0 & SE <= 1)

# Non-Life PTE Distribution (from VRS)
nonlife_pte <- plot_data_all_eff %>%
  filter(Firm_Type == "Non-Life") %>%
  mutate(PTE_Group = cut(PTE,
                         breaks = c(0.6, 0.7, 0.8, 0.9, 0.9999, 1.01),
                         labels = c("0.60 ≤ E < 0.70",
                                    "0.70 ≤ E < 0.80",
                                    "0.80 ≤ E < 0.90",
                                    "0.90 ≤ E < 1.00",
                                    "E = 1.00"),
                         right = FALSE))

table_pte_nonlife <- nonlife_pte %>%
  group_by(PTE_Group) %>%
  summarise(Count = n()) %>%
  mutate(Percent = round(Count / sum(Count) * 100, 1))

kable(table_pte_nonlife,
      caption = "Table 4.6: Distribution of PTE Scores for Non-Life Insurance
Firms",
      col.names = c("Efficiency Interval", "Number of Firms", "Percentage
(%)"))
```

Table 4.6: Distribution of PTE Scores for Non-Life Insurance Firms

Efficiency Interval	Number of Firms	Percentage (%)
0.60 ≤ E < 0.70	4	3.6
0.70 ≤ E < 0.80	8	7.1
0.80 ≤ E < 0.90	38	33.9
0.90 ≤ E < 1.00	32	28.6
E = 1.00	30	26.8

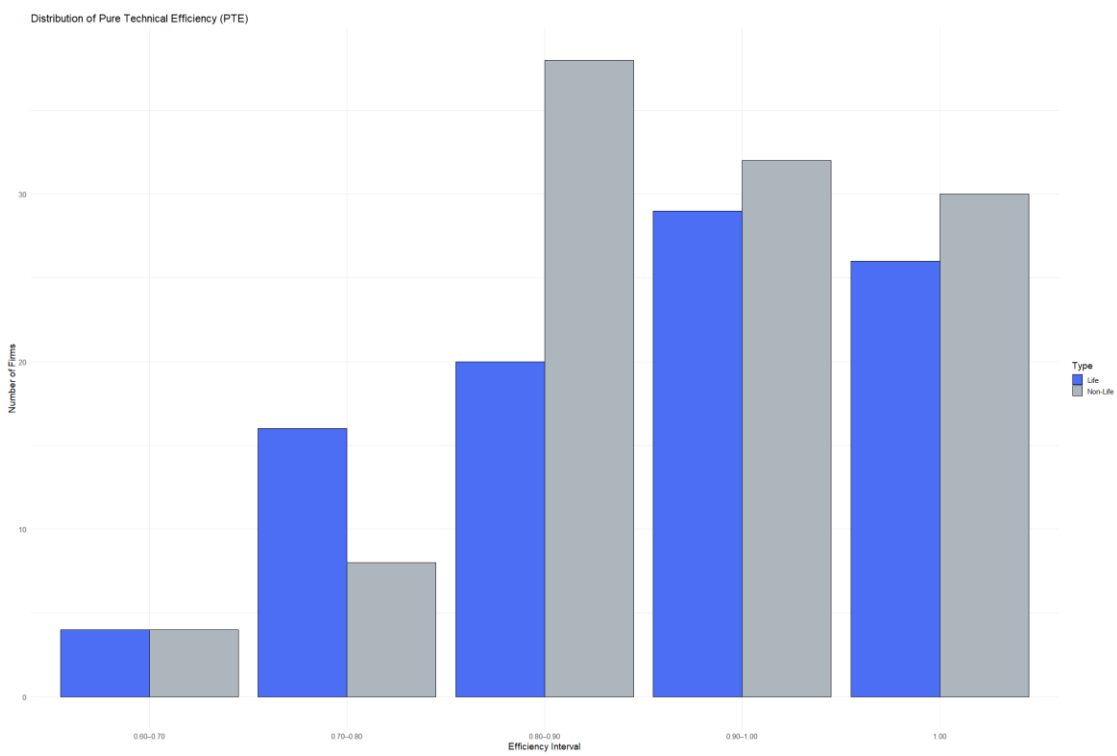
```
# Life PTE Distribution (from VRS)
life_pte <- plot_data_all_eff %>%
  filter(Firm_Type == "Life") %>%
  mutate(PTE_Group = cut(PTE,
                         breaks = c(0.6, 0.7, 0.8, 0.9, 0.9999, 1.01),
                         labels = c("0.60 ≤ E < 0.70",
                                    "0.70 ≤ E < 0.80",
                                    "0.80 ≤ E < 0.90",
                                    "0.90 ≤ E < 1.00",
                                    "E = 1.00"),
                         right = FALSE))

table_pte_life <- life_pte %>%
  group_by(PTE_Group) %>%
  summarise(Count = n()) %>%
  mutate(Percent = round(Count / sum(Count) * 100, 1))

kable(table_pte_life,
      caption = "Table 4.5: Distribution of PTE Scores for Life Insurance
Firms",
      col.names = c("Efficiency Interval", "Number of Firms", "Percentage
(%)"))
```

Table 4.5: Distribution of PTE Scores for Life Insurance Firms

Efficiency Interval	Number of Firms	Percentage (%)
0.60 ≤ E < 0.70	4	4.1
0.70 ≤ E < 0.80	16	16.3
0.80 ≤ E < 0.90	20	20.4
0.90 ≤ E < 1.00	29	29.6
E = 1.00	26	26.5
NA	3	3.1

```
library(ggplot2)
library(dplyr)



life_pte <- data.frame(
```

```r
  Interval = c("0.60–0.70", "0.70–0.80", "0.80–0.90", "0.90–1.00", "1.00"),
  Count = c(4, 16, 20, 29, 26)
)
nonlife_pte <- data.frame(
  Interval = c("0.60–0.70", "0.70–0.80", "0.80–0.90", "0.90–1.00", "1.00"),
  Count = c(4, 8, 38, 32, 30)
)
pte_data <- bind_rows(
  mutate(life_pte, Type = "Life"),
  mutate(nonlife_pte, Type = "Non-Life")
)


ggplot(pte_data, aes(x = Interval, y = Count, fill = Type)) +
  geom_bar(stat = "identity", position = "dodge", color = "black") +
  labs(title = "Distribution of Pure Technical Efficiency (PTE)",
       x = "Efficiency Interval",
       y = "Number of Firms") +
  scale_fill_manual(values = c("Life" = "#4C6EF5", "Non-Life" = "#ADB5BD")) +
  theme_minimal(base_size = 14)
```




```r
#choose variables input/output
dea_vars <- dea_clean %>%
  select(`Total Operating Exp`, `Total Debt`, `Total Equity`,
         `Financial Investment Income`, `Benefits/ losses incurred`) / 1e9
```

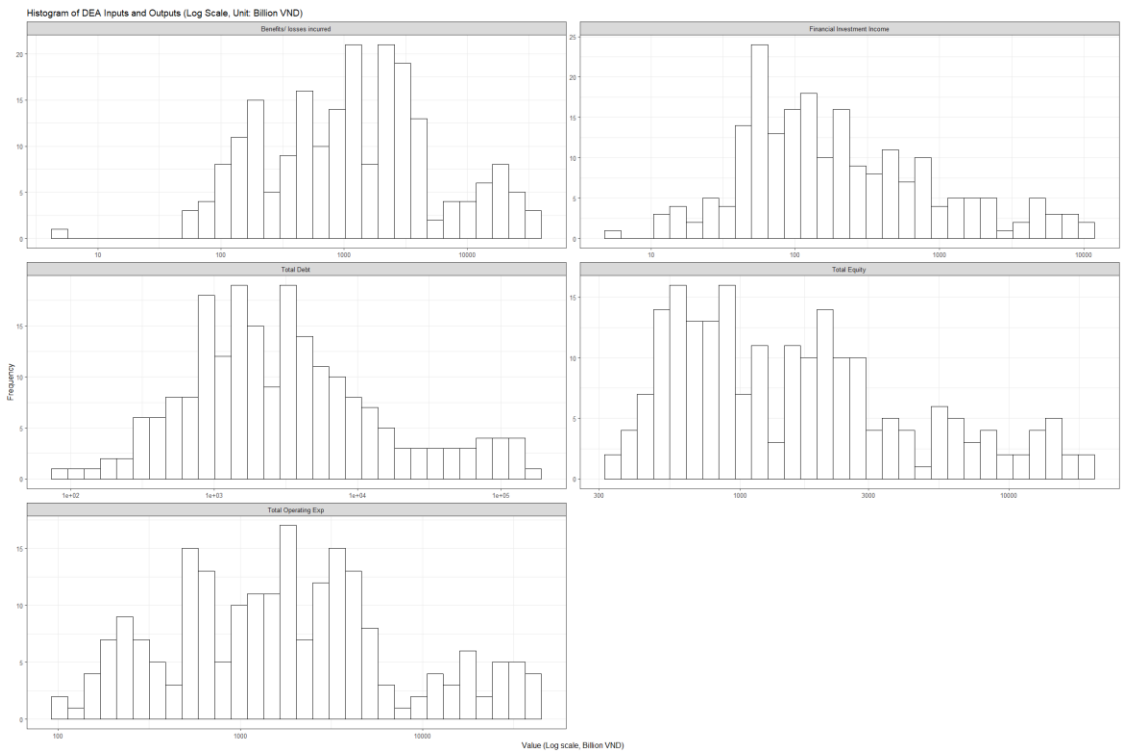
```r
dea_desc_stats <- describe(dea_vars)


dea_table <- round(dea_desc_stats[, c("mean", "sd", "min", "max")], 2)


rownames(dea_table) <- c("Total Operating Expenses",
                         "Total Debt",
                         "Total Equity",
                         "Financial Inv. Income",
                         "Benefits/Losses Incurred")



kable(dea_table,
      caption = "Table 4.1: Descriptive Statistics for DEA Inputs and Outputs
(Unit: Billion VND)",
      digits = 2,
      format.args = list(big.mark = ","))
```

Table 4.1: Descriptive Statistics for DEA Inputs and Outputs (Unit: Billion VND)

	mean	sd	min	max
Total Operating Expenses	4,846.53	8,304.18	101.99	40,472.55
Total Debt	12,137.13	27,258.96	83.57	167,946.98
Total Equity	2,865.60	3,769.40	336.17	19,370.12
Financial Inv. Income	761.80	1,697.94	5.43	10,415.07
Benefits/Losses Incurred	3,633.63	6,490.83	4.89	34,277.43

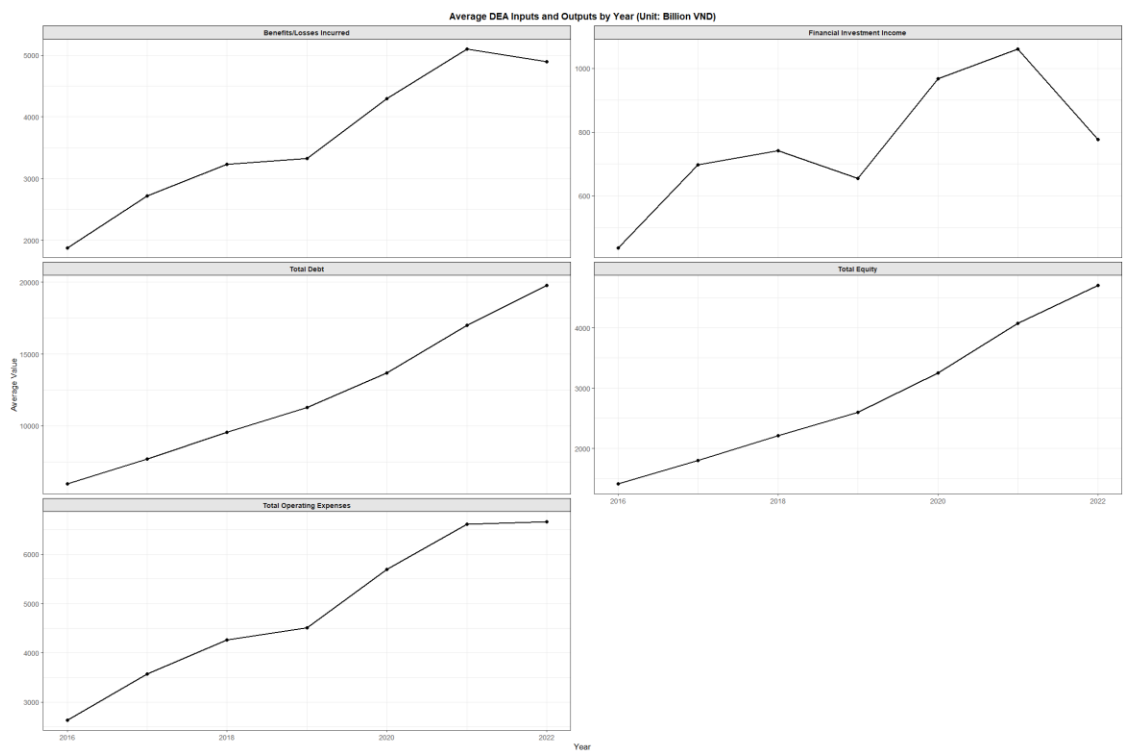
```r
names(dea_clean)

##  [1] "No."                          "Firm"
##  [3] "Life insurance?"              "Year"
##  [5] "DEA Score"                    "Financial Investment Income"
##  [7] "Benefits/ losses incurred"    "Total Equity"
##  [9] "Total Debt"                   "Total Operating Exp"
## [11] "Dummy (Life_Insurance=1)"     "Firm_Type"

dea_clean_billion <- dea_clean %>%
  mutate(
    `Total Operating Exp` = `Total Operating Exp` / 1e9,
    `Total Debt` = `Total Debt` / 1e9,
    `Total Equity` = `Total Equity` / 1e9,
    `Financial Investment Income` = `Financial Investment Income` / 1e9,
    `Benefits/ losses incurred` = `Benefits/ losses incurred` / 1e9
  )

dea_long <- dea_clean_billion %>%
  select(`Total Operating Exp`, `Total Debt`, `Total Equity`,
         `Financial Investment Income`, `Benefits/ losses incurred`) %>%
```

```r
  pivot_longer(cols = everything(), names_to = "Variable", values_to =
"Value")

ggplot(dea_long, aes(x = Value)) +
  geom_histogram(fill = "white", color = "black", bins = 30) +
  facet_wrap(~ Variable, scales = "free", ncol = 2) +
  scale_x_log10() +  # 🌟 chuyển trục X sang log-scale
  theme_bw(base_size = 12) +
  labs(
    title = "Histogram of DEA Inputs and Outputs (Log Scale, Unit: Billion
VND)",
    x = "Value (Log scale, Billion VND)", y = "Frequency"
  )
```




```r
# Average DEA Inputs and Outputs by Year
dea_by_year <- dea_clean %>%
  group_by(Year) %>%
  summarise(
    `Total Operating Expenses` = mean(`Total Operating Exp`, na.rm = TRUE) /
1e9,
    `Total Debt` = mean(`Total Debt`, na.rm = TRUE) / 1e9,
    `Total Equity` = mean(`Total Equity`, na.rm = TRUE) / 1e9,
    `Financial Investment Income` = mean(`Financial Investment Income`, na.rm
= TRUE) / 1e9,
    `Benefits/Losses Incurred` = mean(`Benefits/ losses incurred`, na.rm =
TRUE) / 1e9
  ) %>%
```

```r
  pivot_longer(cols = -Year, names_to = "Variable", values_to = "Avg_Value")


ggplot(dea_by_year, aes(x = Year, y = Avg_Value, group = Variable)) +
  geom_line(color = "black", size = 1) +
  geom_point(color = "black", size = 2) +
  facet_wrap(~ Variable, scales = "free_y", ncol = 2) +
  theme_bw(base_size = 13) +
  labs(
    title = "Average DEA Inputs and Outputs by Year (Unit: Billion VND)",
    x = "Year",
    y = "Average Value"
  ) +
  theme(
    strip.background = element_rect(fill = "gray90", color = "black"),
    strip.text = element_text(face = "bold"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
  )
```



#1.9 Save All Efficiency Scores

```r
# Save the detailed efficiency scores to a CSV file
write.csv(all_eff_scores_detailed,
"D:/dea/deaxl/all_efficiency_scores_detailed.csv", row.names = FALSE)

cat("\n--- All Efficiency Scores with Firm Type Saved to
all_efficiency_scores_detailed.csv ---\n")
```

```
##
## --- All Efficiency Scores with Firm Type Saved to
all_efficiency_scores_detailed.csv ---
```

```r
head(all_eff_scores_detailed, n=10)
```

```
##     Firm Year Firm_Type       OTE       PTE        SE
## 1    AIA 2016      Life 0.7884149 0.7938695 0.9931290
## 2    AIA 2017      Life 0.8086187 0.8098457 0.9984849
## 3    AIA 2018      Life 0.8102293 0.8115008 0.9984332
## 4    AIA 2019      Life 0.7741838 0.7789558 0.9938739
## 5    AIA 2020      Life 0.7812774 0.7865433 0.9933050
## 6    AIA 2021      Life 0.8904749 0.9085229 0.9801348
## 7    AIA 2022      Life 0.8318629 0.8528930 0.9753427
## 8  Aviva 2016      Life 1.0000000 1.0000000 1.0000000
## 9  Aviva 2017      Life 0.8818186 0.8875227 0.9935730
## 10 Aviva 2018      Life 0.8065196 0.8072481 0.9990976
```

```r
table_avg_eff_firmtype <- plot_data_all_eff %>%
  group_by(Firm_Type) %>%
  summarise(
    Avg_OTE = round(mean(OTE, na.rm = TRUE), 3),
    Avg_PTE = round(mean(PTE, na.rm = TRUE), 3),
    Avg_SE = round(mean(SE, na.rm = TRUE), 3)
  )

table_avg_eff_firmtype
```

```
## # A tibble: 2 × 4
##   Firm_Type Avg_OTE Avg_PTE Avg_SE
##   <chr>       <dbl>   <dbl>  <dbl>
## 1 Life        0.858   0.886  0.969
## 2 Non-Life    0.833   0.908  0.918
```

Table Average by Year

```r
table_avg_eff_year <- plot_data_all_eff %>%
  group_by(Year) %>%
  summarise(
    Avg_OTE = round(mean(OTE, na.rm = TRUE), 3),
    Avg_PTE = round(mean(PTE, na.rm = TRUE), 3),
    Avg_SE = round(mean(SE, na.rm = TRUE), 3)
  )

table_avg_eff_year
```

```
## # A tibble: 7 × 4
##    Year Avg_OTE Avg_PTE Avg_SE
##   <dbl>   <dbl>   <dbl>  <dbl>
## 1  2016   0.835   0.904  0.923
```

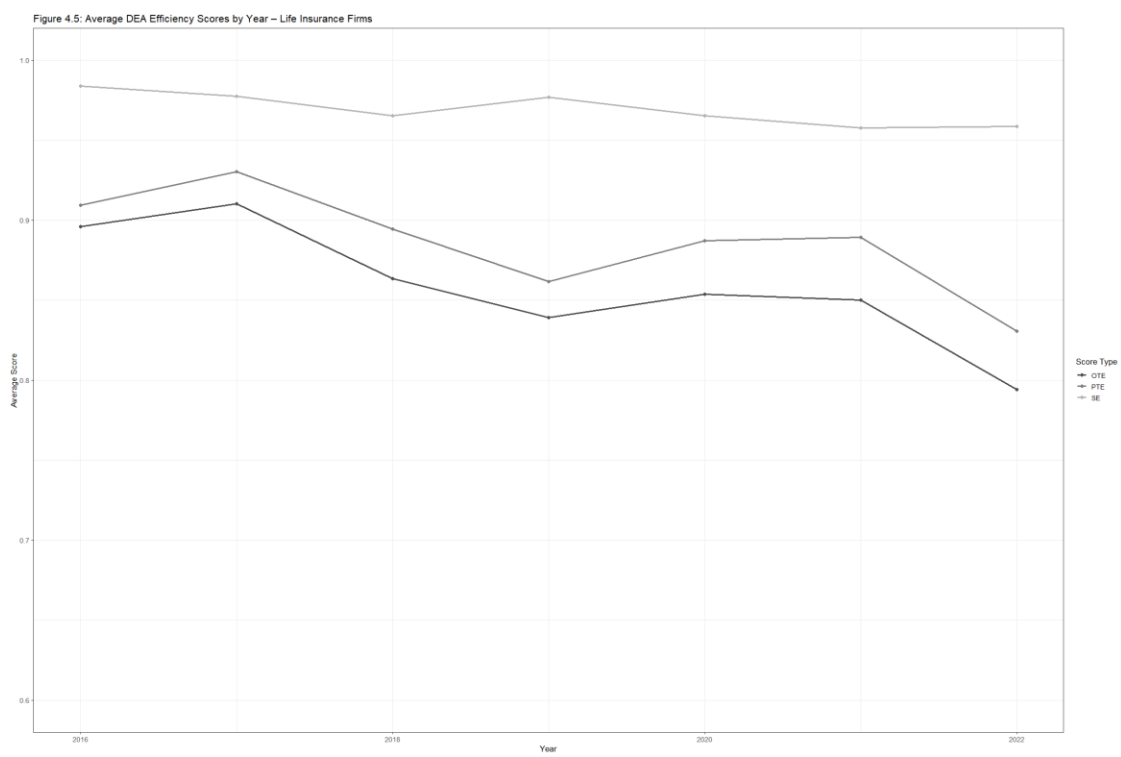
```
## 2  2017   0.859   0.917  0.936
## 3  2018   0.83    0.896  0.928
## 4  2019   0.834   0.881  0.949
## 5  2020   0.868   0.909  0.957
## 6  2021   0.847   0.897  0.946
## 7  2022   0.839   0.88   0.953
```

```
 #Calculate descriptive statistics for DEA indices
dea_stats <- all_eff_scores_detailed %>%
  select(OTE, PTE, SE) %>%
  summarise(
    OTE_Mean = mean(OTE, na.rm = TRUE),
    OTE_SD   = sd(OTE, na.rm = TRUE),
    OTE_Min  = min(OTE, na.rm = TRUE),
    OTE_Max  = max(OTE, na.rm = TRUE),

    PTE_Mean = mean(PTE, na.rm = TRUE),
    PTE_SD   = sd(PTE, na.rm = TRUE),
    PTE_Min  = min(PTE, na.rm = TRUE),
    PTE_Max  = max(PTE, na.rm = TRUE),

    SE_Mean = mean(SE, na.rm = TRUE),
    SE_SD   = sd(SE, na.rm = TRUE),
    SE_Min  = min(SE, na.rm = TRUE),
    SE_Max  = max(SE, na.rm = TRUE)
  )

dea_summary <- tibble::tibble(
  `Score Type` = c("OTE", "PTE", "SE"),
  Mean     = c(dea_stats$OTE_Mean, dea_stats$PTE_Mean, dea_stats$SE_Mean),
  `Std. Dev.` = c(dea_stats$OTE_SD, dea_stats$PTE_SD, dea_stats$SE_SD),
  Min      = c(dea_stats$OTE_Min, dea_stats$PTE_Min, dea_stats$SE_Min),
  Max      = c(dea_stats$OTE_Max, dea_stats$PTE_Max, dea_stats$SE_Max)
)

dea_summary[, 2:5] <- round(dea_summary[, 2:5], 2)

dea_summary
```

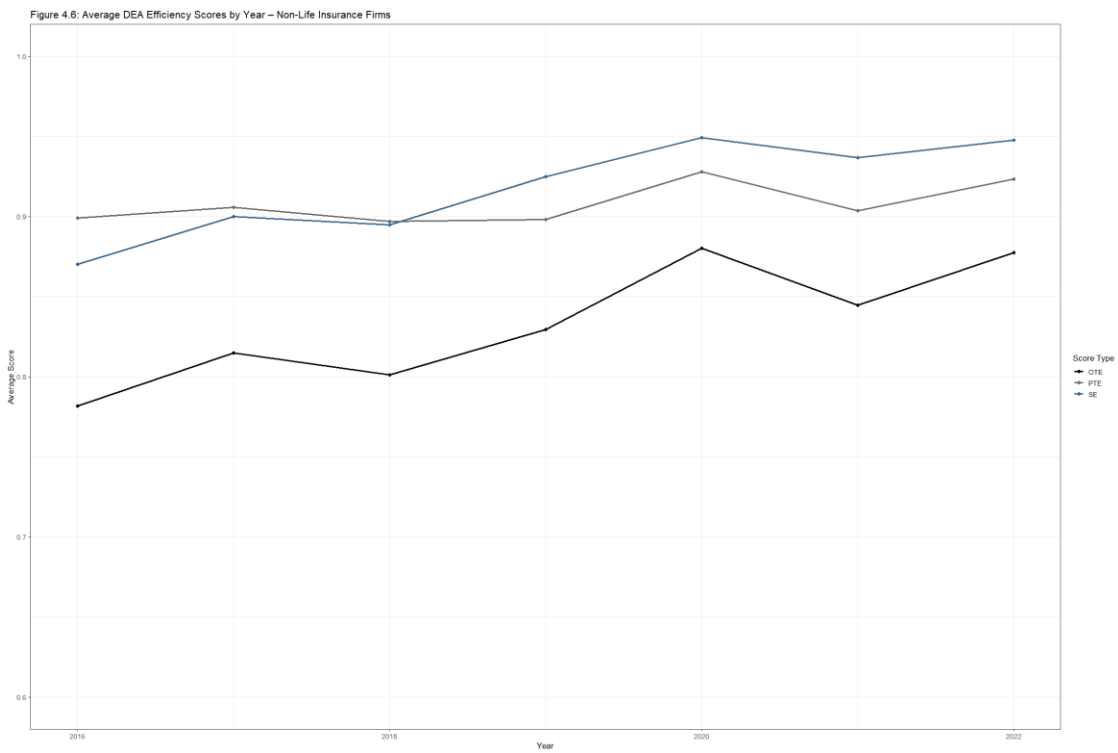
```
## # A tibble: 3 × 5
##   `Score Type`  Mean `Std. Dev.`   Min   Max
##   <chr>        <dbl>       <dbl> <dbl> <dbl>
## 1 OTE           0.84        0.12  0.5      1
## 2 PTE           0.9         0.1   0.53     1
## 3 SE            0.94        0.08  0.66     1
```

```
# Average DEA Efficiency Scores by Year - Life Insurance Firms
eff_by_year_life <- all_eff_scores_detailed %>%
  filter(Firm_Type == "Life") %>%
  group_by(Year) %>%
```

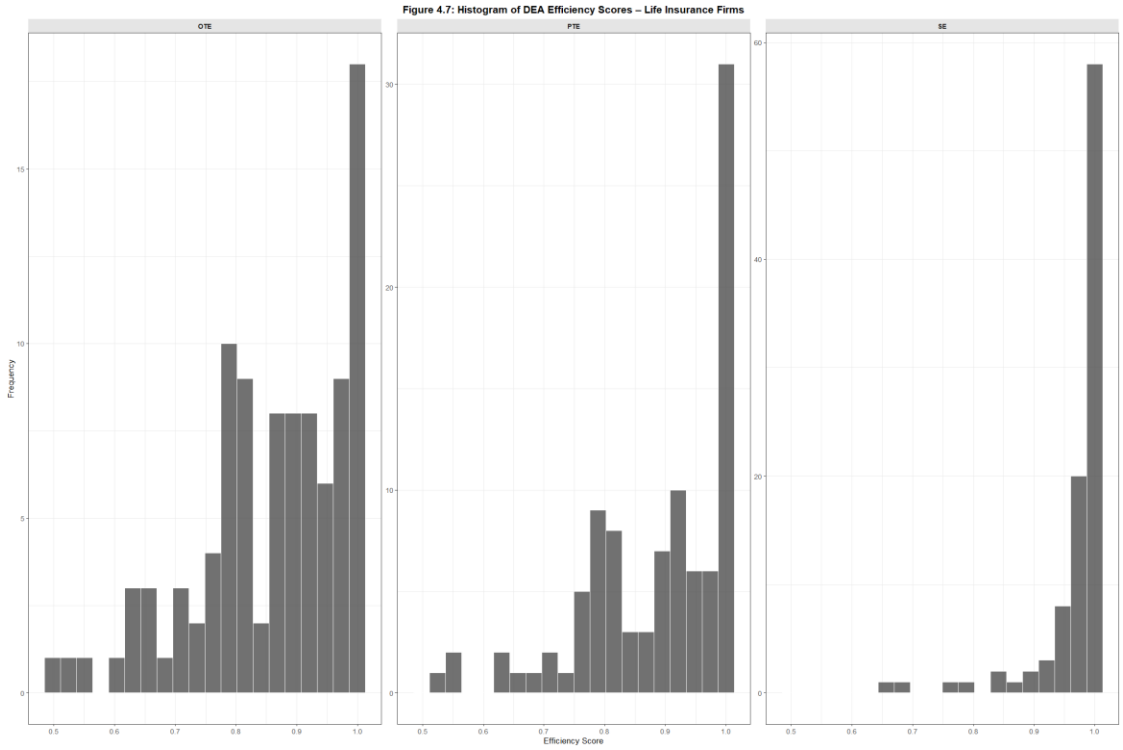
```r
  summarise(
    Avg_OTE = mean(OTE, na.rm = TRUE),
    Avg_PTE = mean(PTE, na.rm = TRUE),
    Avg_SE = mean(SE, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = starts_with("Avg_"), names_to = "Score_Type", values_to
= "Value")

ggplot(eff_by_year_life, aes(x = Year, y = Value, color = Score_Type, group =
Score_Type)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  scale_color_manual(
    values = c("Avg_OTE" = "gray30",
               "Avg_PTE" = "gray50",
               "Avg_SE"  = "gray70"),
    labels = c("OTE", "PTE", "SE")
  ) +
  labs(
    title = "Figure 4.5: Average DEA Efficiency Scores by Year – Life
Insurance Firms",
    x = "Year", y = "Average Score", color = "Score Type"
  ) +
  theme_bw(base_size = 13) +
  ylim(0.6, 1.0)
```



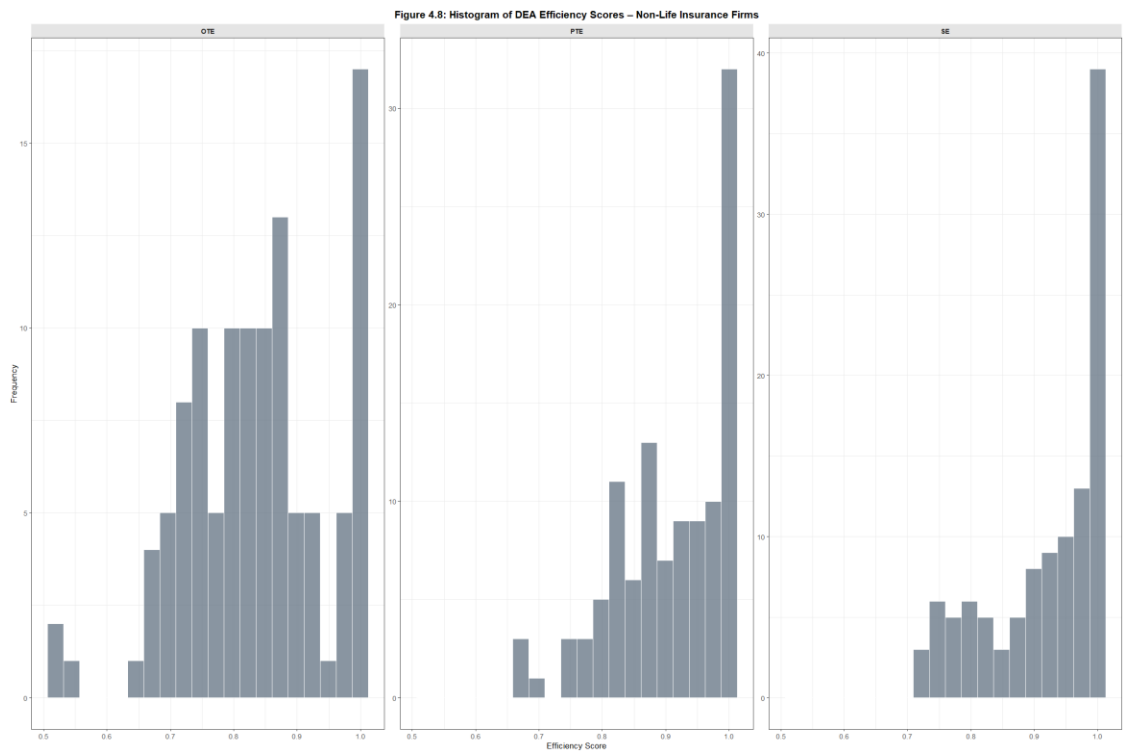
```r
# Average DEA Efficiency Scores by Year – Non-Life Insurance Firms
eff_by_year_nonlife <- all_eff_scores_detailed %>%
  filter(Firm_Type == "Non-Life") %>%
  group_by(Year) %>%
  summarise(
    Avg_OTE = mean(OTE, na.rm = TRUE),
    Avg_PTE = mean(PTE, na.rm = TRUE),
    Avg_SE = mean(SE, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = starts_with("Avg_"), names_to = "Score_Type", values_to
= "Value")


ggplot(eff_by_year_nonlife, aes(x = Year, y = Value, color = Score_Type,
group = Score_Type)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  scale_color_manual(
    values = c("Avg_OTE" = "black",
               "Avg_PTE" = "dimgray",
               "Avg_SE"  = "steelblue4"),
    labels = c("OTE", "PTE", "SE")
  ) +
  labs(
    title = "Figure 4.6: Average DEA Efficiency Scores by Year – Non-Life
Insurance Firms",
    x = "Year", y = "Average Score", color = "Score Type"
  ) +
  theme_bw(base_size = 13) +
  ylim(0.6, 1.0)
```




```
# Convert to long form for easy histogram drawing
eff_hist_data <- all_eff_scores_detailed %>%
  pivot_longer(cols = c(OTE, PTE, SE),
               names_to = "Efficiency_Type",
               values_to = "Efficiency_Score") %>%
  filter(!is.na(Efficiency_Score))

# Histogram for Life Firms
eff_hist_data %>%
  filter(Firm_Type == "Life") %>%
  ggplot(aes(x = Efficiency_Score)) +
  geom_histogram(bins = 20, fill = "#4D4D4D", color = "white", alpha = 0.8) +
  facet_wrap(~ Efficiency_Type, scales = "free_y") +
  theme_bw(base_size = 13) +
  labs(
    title = "Figure 4.7: Histogram of DEA Efficiency Scores – Life Insurance
Firms",
    x = "Efficiency Score",
    y = "Frequency"
  ) +
  theme(strip.background = element_rect(fill = "gray90", color = NA),
        strip.text = element_text(face = "bold"),
        plot.title = element_text(hjust = 0.5, face = "bold"))
```




```r
# Histogram for Non-Life Firms
eff_hist_data %>%
  filter(Firm_Type == "Non-Life") %>%
  ggplot(aes(x = Efficiency_Score)) +
  geom_histogram(bins = 20, fill = "#6C7A89", color = "white", alpha = 0.8) +
  facet_wrap(~ Efficiency_Type, scales = "free_y") +
  theme_bw(base_size = 13) +
  labs(
    title = "Figure 4.8: Histogram of DEA Efficiency Scores – Non-Life
Insurance Firms",
    x = "Efficiency Score",
    y = "Frequency"
  ) +
  theme(strip.background = element_rect(fill = "gray90", color = NA),
        strip.text = element_text(face = "bold"),
        plot.title = element_text(hjust = 0.5, face = "bold"))
```



Phase 2: Determinants Analysis using Truncated Regression

This stage combines the DEA efficiency scores with firm-specific and macroeconomic determinants and performs truncated regression analysis. The use of a truncated regression model is appropriate given the bounded nature of DEA efficiency scores between 0 and 1.

#2.1 Data Import and Initial Cleaning

```
reg_raw <- read_excel("D:/data/regress/Regression.xlsx", sheet = "Sheet1")

cat("--- Raw Regression Data (reg_raw) ---\n")

## --- Raw Regression Data (reg_raw) ---

head(reg_raw, n=10)

## # A tibble: 10 × 9
##    Firm   Year   LEV    GWP  SIZE   AGE LIFEDUM `Interest Rate`
`%Inflation`
##    <chr> <dbl> <dbl>  <dbl> <dbl> <dbl>   <dbl>           <dbl>
<dbl>
##  1 AIA    2016  5.79 0.351   29.9  3.04       1          0.0610        3.5
##  2 AIA    2017  3.15 0.322   30.4  3.04       1          0.0505
0.296
##  3 AIA    2018  3.50 0.350   30.6  3.04       1          0.0517        0
```

```
##  4 AIA    2019  3.76 0.325   30.8  3.04       1          0.0333       -0.2
##  5 AIA    2020  3.64 0.257   31.0  3.04       1          0.0228
0.143
##  6 AIA    2021  4.32 0.160   31.3  3.04       1          0.0205       -
0.438
##  7 AIA    2022  2.93 0.113   13.7  3.04       1          0.0504
0.727
##  8 ABIC   2016  1.57 0.322   28.1  2.64       0          0.0610        3.5
##  9 ABIC   2017  1.72 0.0853  28.2  2.64       0          0.0505
0.296
## 10 ABIC   2018  1.82 0.176   28.4  2.64       0          0.0517        0
```

```r
str(reg_raw)
```

```
## tibble [210 × 9] (S3: tbl_df/tbl/data.frame)
##  $ Firm         : chr [1:210] "AIA" "AIA" "AIA" "AIA" ...
##  $ Year         : num [1:210] 2016 2017 2018 2019 2020 ...
##  $ LEV          : num [1:210] 5.79 3.15 3.5 3.76 3.64 ...
##  $ GWP          : num [1:210] 0.351 0.322 0.35 0.325 0.257 ...
##  $ SIZE         : num [1:210] 29.9 30.4 30.6 30.8 31 ...
##  $ AGE          : num [1:210] 3.04 3.04 3.04 3.04 3.04 ...
##  $ LIFEDUM      : num [1:210] 1 1 1 1 1 1 1 0 0 0 ...
##  $ Interest Rate: num [1:210] 0.0609 0.0505 0.0517 0.0333 0.0228 ...
##  $ %Inflation   : num [1:210] 3.5 0.296 0 -0.2 0.143 ...
```

#2.2 Clean and Transform Regression Variables

```r
# Clean and prepare regression variables
reg_clean <- reg_raw %>%
  filter(!is.na(LEV), !is.na(GWP), !is.na(SIZE), !is.na(`Interest Rate`),
!is.na(`%Inflation`)) %>%
  mutate(
    across(c(LEV, GWP, SIZE, AGE, LIFEDUM, `Interest Rate`, `%Inflation`),
as.numeric),

    Size_log = log(SIZE)
  ) %>%
  rename(Leverage = LEV,
         Gross_Written_Premium = GWP,
         Interest_Rate = `Interest Rate`, # Macroeconomic variable
         Inflation_Rate = `%Inflation`) %>% # Macroeconomic variable
  group_by(Firm) %>%
  mutate(Growth = (Gross_Written_Premium - lag(Gross_Written_Premium)) /
lag(Gross_Written_Premium)) %>%
  ungroup() %>%
  filter(!is.na(Growth))

eff_data <- read.csv("D:/dea/deaxl/all_efficiency_scores_detailed.csv")
```

#2.3 Merge Regression Data with DEA Efficiency Scores

```r
reg_merged <- left_join(reg_clean, eff_data, by = c("Firm", "Year"))

reg_merged <- reg_merged %>%
  filter(!is.na(OTE),
         !is.na(Leverage),
         !is.na(Gross_Written_Premium),
         !is.na(Size_log),
         !is.na(Growth),
         !is.na(Interest_Rate),
         !is.na(Inflation_Rate),
         !is.na(AGE),
         !is.na(Firm_Type)) # Crucial for splitting analysis

cat("\n--- Merged Data for Regression (reg_merged) ---\n")
```

```
##
## --- Merged Data for Regression (reg_merged) ---
```

```r
head(reg_merged)
```

```
## # A tibble: 6 × 15
##   Firm   Year Leverage Gross_Written_Premium  SIZE   AGE LIFEDUM
Interest_Rate
##   <chr> <dbl>    <dbl>                 <dbl> <dbl> <dbl>   <dbl>
<dbl>
## 1 AIA    2017     3.15                 0.322  30.4  3.04       1
0.0505
## 2 AIA    2018     3.50                 0.350  30.6  3.04       1
0.0517
## 3 AIA    2019     3.76                 0.325  30.8  3.04       1
0.0333
## 4 AIA    2020     3.64                 0.257  31.0  3.04       1
0.0228
## 5 AIA    2021     4.32                 0.160  31.3  3.04       1
0.0205
## 6 AIA    2022     2.93                 0.113  13.7  3.04       1
0.0504
## # ℹ 7 more variables: Inflation_Rate <dbl>, Size_log <dbl>, Growth
<dbl>,
## #   Firm_Type <chr>, OTE <dbl>, PTE <dbl>, SE <dbl>
```

```r
str(reg_merged)
```

```
## tibble [180 × 15] (S3: tbl_df/tbl/data.frame)
##  $ Firm                 : chr [1:180] "AIA" "AIA" "AIA" "AIA" ...
##  $ Year                 : num [1:180] 2017 2018 2019 2020 2021 ...
##  $ Leverage             : num [1:180] 3.15 3.5 3.76 3.64 4.32 ...
##  $ Gross_Written_Premium: num [1:180] 0.322 0.35 0.325 0.257 0.16 ...
##  $ SIZE                 : num [1:180] 30.4 30.6 30.8 31 31.3 ...
##  $ AGE                  : num [1:180] 3.04 3.04 3.04 3.04 3.04 ...
##  $ LIFEDUM              : num [1:180] 1 1 1 1 1 1 0 0 0 0 ...
```

```
##  $ Interest_Rate        : num [1:180] 0.0505 0.0517 0.0333 0.0228 0.0205
...
##  $ Inflation_Rate       : num [1:180] 0.296 0 -0.2 0.143 -0.438 ...
##  $ Size_log             : num [1:180] 3.41 3.42 3.43 3.44 3.44 ...
##  $ Growth               : num [1:180] -0.083 0.0859 -0.0696 -0.2109 -
0.3768 ...
##  $ Firm_Type            : chr [1:180] "Life" "Life" "Life" "Life" ...
##  $ OTE                  : num [1:180] 0.809 0.81 0.774 0.781 0.89 ...
##  $ PTE                  : num [1:180] 0.81 0.812 0.779 0.787 0.909 ...
##  $ SE                   : num [1:180] 0.998 0.998 0.994 0.993 0.98 ...
```

```r
summary(reg_merged$OTE)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.5189  0.7762  0.8555  0.8461  0.9435  1.0000
```

#2.4 Descriptive Statistics of Regression Variables

```r
reg_vars <- reg_merged %>%
  select(OTE, Size_log, Leverage, Growth, Interest_Rate, Inflation_Rate, AGE)

desc_stats <- describe(reg_vars)
summary_table <- round(desc_stats[, c("mean", "sd", "min", "max")], 2)

rownames(summary_table) <- c("OTE", "Size_log", "Leverage", "Growth",
                             "Interest_Rate", "Inflation_Rate", "AGE")

kable(summary_table,
      caption = "Table 4.2: Descriptive Statistics for Truncated Regression",
      col.names = c("Mean", "Std. Dev.", "Min", "Max"))
```

Table 4.2: Descriptive Statistics for Truncated Regression

	Mean	Std. Dev.	Min	Max
OTE	0.85	0.12	0.52	1.00
Size_log	3.24	0.31	2.49	3.49
Leverage	2.83	3.43	0.14	20.27
Growth	2.33	22.72	-33.60	287.75
Interest_Rate	0.04	0.01	0.02	0.05
Inflation_Rate	0.09	0.37	-0.44	0.73
AGE	2.81	0.44	1.95	4.03

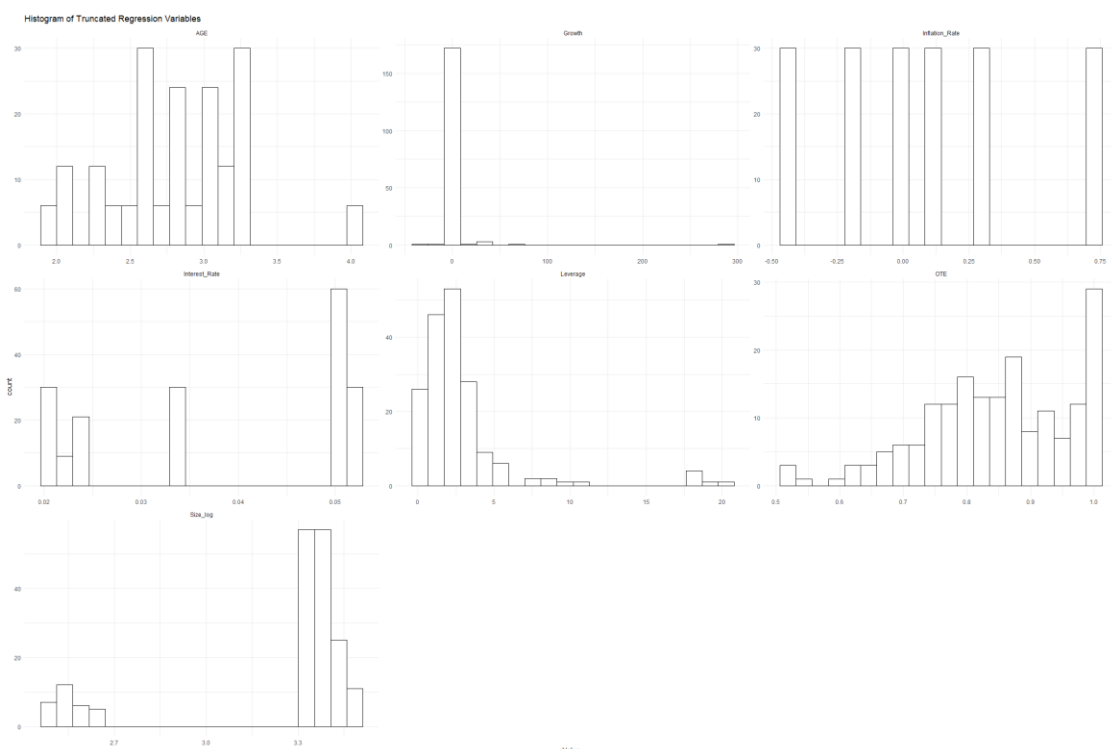
#2.5 Subset Data by Firm Type (Life vs Non-Life)

```r
life_data <- reg_merged %>% filter(Firm_Type == "Life")
nonlife_data <- reg_merged %>% filter(Firm_Type == "Non-Life")

cat("Life Insurance Firms:", nrow(life_data), "\n")
```

```
## Life Insurance Firms: 84

cat("Non-Life Insurance Firms:", nrow(nonlife_data), "\n")

## Non-Life Insurance Firms: 96
```

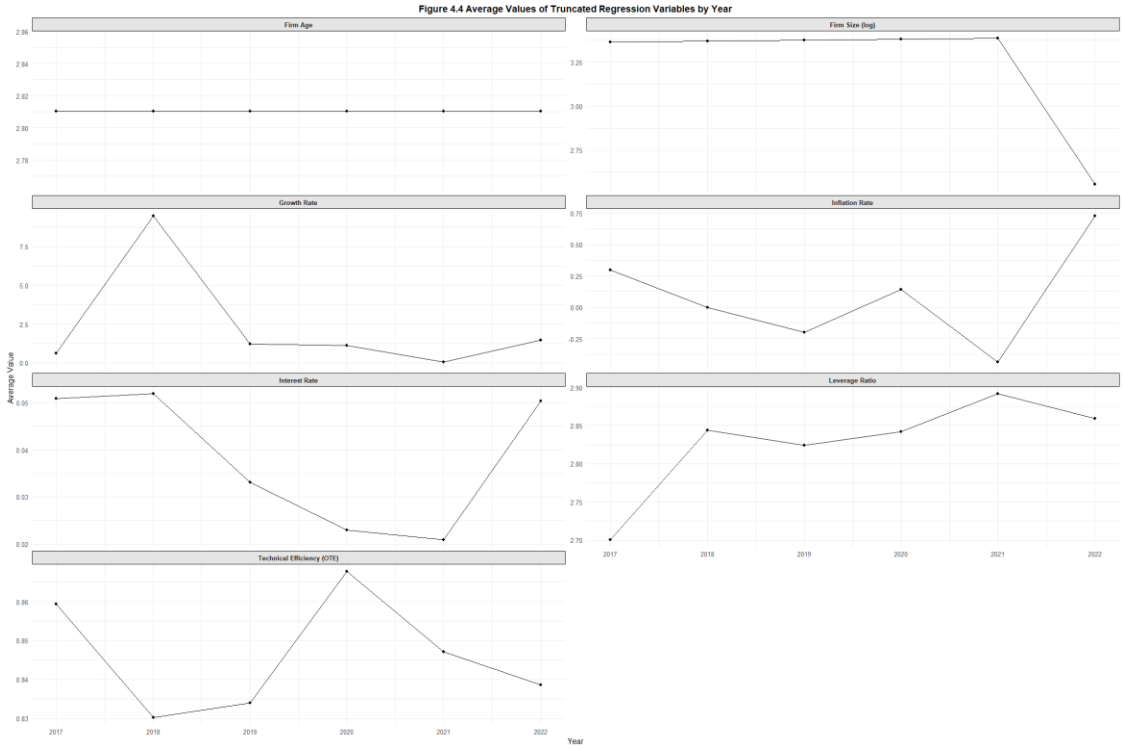
#2.6 Visualize Regression Variables (Histograms)

```
library(gridExtra)



reg_vars_long <- reg_vars %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to =
"Value")

ggplot(reg_vars_long, aes(x = Value)) +
  geom_histogram(color = "black", fill = "white", bins = 20) +
  facet_wrap(~ Variable, scales = "free", ncol = 3) +
  labs(title = "Histogram of Truncated Regression Variables", x = "Value") +
  theme_minimal()
```




```
reg_vars_long <- reg_merged %>%
  select(Year, OTE, Size_log, Leverage, Growth, Interest_Rate,
Inflation_Rate, AGE) %>%
  pivot_longer(cols = -Year, names_to = "Variable", values_to = "Value")

reg_data_yearly <- reg_vars_long %>%
```

```r
  group_by(Year, Variable) %>%
  summarise(Average = mean(Value, na.rm = TRUE), .groups = "drop")

reg_data_yearly$Variable <- dplyr::recode(reg_data_yearly$Variable,
                                          "OTE" = "Technical Efficiency (OTE)",
                                          "Size_log" = "Firm Size (log)",
                                          "Leverage" = "Leverage Ratio",
                                          "Growth" = "Growth Rate",
                                          "Interest_Rate" = "Interest Rate",
                                          "Inflation_Rate" = "Inflation Rate",
                                          "AGE" = "Firm Age")

reg_data_yearly$Variable <- case_when(
  reg_data_yearly$Variable == "OTE" ~ "Technical Efficiency (OTE)",
  reg_data_yearly$Variable == "Size_log" ~ "Firm Size (log)",
  reg_data_yearly$Variable == "Leverage" ~ "Leverage Ratio",
  reg_data_yearly$Variable == "Growth" ~ "Growth Rate",
  reg_data_yearly$Variable == "Interest_Rate" ~ "Interest Rate",
  reg_data_yearly$Variable == "Inflation_Rate" ~ "Inflation Rate",
  reg_data_yearly$Variable == "AGE" ~ "Firm Age",
  TRUE ~ reg_data_yearly$Variable
)

ggplot(reg_data_yearly, aes(x = Year, y = Average)) +
  geom_line(color = "black") +
  geom_point(color = "black") +
  facet_wrap(~ Variable, scales = "free_y", ncol = 2) +
  labs(title = "Figure 4.4 Average Values of Truncated Regression Variables
by Year",
       x = "Year", y = "Average Value") +
  theme_minimal(base_size = 12) +
  theme(strip.background = element_rect(fill = "gray90"),
        strip.text = element_text(face = "bold"),
        plot.title = element_text(hjust = 0.5, face = "bold"))
```



#2.7 Run Truncated Regression Models:

```
# Truncated Regression For All Insurance Firm
trunc_all <- truncreg(OTE ~ Leverage + Size_log + AGE + Growth +
Interest_Rate + Inflation_Rate,
                      data = reg_merged,
                      lower = 0, # Giới hạn dưới của biến phụ thuộc
                      upper = 1) # Giới hạn trên của biến phụ thuộc

screenreg(trunc_all,
          custom.coef.names = c("(Intercept)", "Leverage", "Log of Size",
"Firm Age", "Growth", "Interest Rate", "Inflation Rate", "Sigma"), # Đã thêm
"Sigma"
          custom.model.names = "All Insurance Firms",
          caption = "Table 4.8: Truncated Regression Results for All
Insurance Firms",
          digits = 3)

##
## ===================================
##                 All Insurance Firms
## -----------------------------------
## (Intercept)        0.525 ***
##                   (0.146)
## Leverage           0.006 *
##                   (0.003)
## Log of Size        0.062
##                   (0.042)
```

```
## Firm Age           0.048 *
##                   (0.021)
## Growth             0.000
##                   (0.000)
## Interest Rate     -1.053
##                   (0.805)
## Inflation Rate     0.069
##                   (0.041)
## Sigma              0.108 ***
##                   (0.006)
## -----------------------------------
## Num. obs.        180
## Log Likelihood   145.670
## AIC             -275.339
## BIC             -249.796
## ===================================
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

```
# Truncated Regression For Life Insurance Firms
trunc_life <- truncreg(OTE ~ Leverage + Size_log + AGE + Growth +
Interest_Rate + Inflation_Rate,
                       data = life_data,
                       lower = 0,
                       upper = 1)
summary(trunc_life)
```

```
##
## Call:
## truncreg(formula = OTE ~ Leverage + Size_log + AGE + Growth +
##     Interest_Rate + Inflation_Rate, data = life_data, lower = 0,
##     upper = 1)
##
## BFGS maximization method
## 51 iterations, 0h:0m:0s
## g'(-H)^-1g = 6.48E-09
##
##
##
## Coefficients :
##                 Estimate Std. Error t-value Pr(>|t|)
## (Intercept)    0.2730173  0.2128144  1.2829  0.19953
## Leverage       0.0080015  0.0039307  2.0357  0.04179 *
## Size_log       0.1474141  0.0631353  2.3349  0.01955 *
## AGE            0.0103488  0.0351747  0.2942  0.76860
## Growth         0.0027718  0.0060933  0.4549  0.64918
## Interest_Rate  0.8987474  1.1727206  0.7664  0.44345
## Inflation_Rate 0.0514386  0.0609627  0.8438  0.39880
## sigma          0.1066956  0.0082317 12.9616  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Log-Likelihood: 68.782 on 8 Df
```

```r
# Truncated Regression For Non-Life Insurance Firms
trunc_nonlife <- truncreg(OTE ~ Leverage + Size_log + AGE + Growth +
Interest_Rate + Inflation_Rate,
                          data = nonlife_data,
                          lower = 0,
                          upper = 1)
summary(trunc_nonlife)
```

```
##
## Call:
## truncreg(formula = OTE ~ Leverage + Size_log + AGE + Growth +
##     Interest_Rate + Inflation_Rate, data = nonlife_data, lower = 0,
##     upper = 1)
##
## BFGS maximization method
## 85 iterations, 0h:0m:0s
## g'(-H)^-1g = 6.16E-07
##
##
##
## Coefficients :
##                    Estimate  Std. Error t-value  Pr(>|t|)
## (Intercept)     0.57190110  0.19645034  2.9112 0.0036007 **
## Leverage        0.01454555  0.01334323  1.0901 0.2756661
## Size_log       -0.00748640  0.05261182 -0.1423 0.8868470
## AGE             0.12248528  0.03670131  3.3374 0.0008458 ***
## Growth          0.00031166  0.00033391  0.9334 0.3506361
## Interest_Rate  -2.69791962  1.02042590 -2.6439 0.0081953 **
## Inflation_Rate  0.08838184  0.05228201  1.6905 0.0909356 .
## sigma           0.09906763  0.00715019 13.8553 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: 85.735 on 8 Df
```

#2.8 Summary Table

```r
screenreg(list("Life Insurance" = trunc_life, "Non-Life Insurance" =
trunc_nonlife),
          custom.model.names = c("Life Insurance", "Non-Life Insurance"),
          digits = 3,
          caption = "Truncated Regression Results for Life and Non-Life
Insurance Firms")
```

```
##
## ==================================================
##                  Life Insurance  Non-Life Insurance
## --------------------------------------------------
```

```
## (Intercept)        0.273           0.572 **
##                   (0.213)         (0.196)
## Leverage           0.008 *         0.015
##                   (0.004)         (0.013)
## Size_log           0.147 *        -0.007
##                   (0.063)         (0.053)
## AGE                0.010           0.122 ***
##                   (0.035)         (0.037)
## Growth             0.003           0.000
##                   (0.006)         (0.000)
## Interest_Rate      0.899          -2.698 **
##                   (1.173)         (1.020)
## Inflation_Rate     0.051           0.088
##                   (0.061)         (0.052)
## sigma              0.107 ***       0.099 ***
##                   (0.008)         (0.007)
## ---------------------------------------------------
## Num. obs.         84              96
## Log Likelihood    68.782          85.735
## AIC             -121.564        -155.469
## BIC             -102.117        -134.955
## ===================================================
## *** p < 0.001; ** p < 0.01; * p < 0.05
```